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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCBTG100 – APPLIED BIOTECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall Outcome attainability.

Part A-One Word Answer

Answer any 10 questions-Each question carries 2 marks

1. What is a potential challenge of implementing personalized medicine? [U] / [CO1]
   Any one challenge- 2 marks.
2. Explain the mechanism of metal recovery in biohydrometallurgy. [U] / [CO2]
   Any mechanism with biological agents-2 marks.
3. What is the importance of CRISPR-Cas9 technology? [K] / [CO4]
   Any one importance-2 marks.
4. List out the microorganisms used for SCP production. [U] / [CO2]
   Any two microorganisms-2 marks.
5. Define transgenic animals? [K] / [CO3]
   Definition- 2 marks.
6. Distinguish between biofuels and fossil fuels in terms of sustainability. [U] / [CO1]
   Biofuels – 1 mark, fossil fuels- 1 mark.
7. Memorize xenograft. [K] / [CO4]
   Xenograft – definition- 2 marks.
8. What is the primary goal of waste management? [U] / [CO2]
   Explain the primary goal- 2 mark.
9. Define GMO? [K] / [CO3].
   GMO- expansion- 1 mark, definition- 1 mark.
10. What is ELSI? [K] / [CO4]
    ELSI- expansion- 2 marks.

11. What is the major tools in recombinant DNA technology? [K] / [CO4]
    Major tools- (minimum two)- 2 marks.
12. Define bio-entrepreneurship [U] / [CO3]
    Definition- 2 marks [2x10 = 20]

Part B-Short Essay

Answer any 5 questions-Each question carries 4 marks

13. What are the ethical considerations for emerging Biotechnology? [K] / [CO1]
    Emerging Biotechnology- ethical considerations (4points)- 4 marks.
14. Discuss the objectives of Human Genome Project. [U] / [CO4].
    4 objectives- 4 marks.
15. What is DBT? Mention it's role. [U] / [CO3]
    DBT- expansion- 1 mark, Role- 3 points- 3 marks.
16. Discuss how biotechnology techniques can be used in forensic science. [U] / [CO4].
    Any two techniques- 2 marks, explain any one- 2 marks.
17. Clarify the concept of selective breeding through Artificial insemination. [U] / [CO3]
    Artificial insemination-definition- 1 mark, selective breeding through Artificial insemination-explanation- 3 marks.
18. Elucidate the main goals of using nanobiotechnology in diagnosis and treatment of diseases? [U] / [CO4]
    Nanobiotechnology- definition-1 mark, role in diagnosis and treatment of diseases-3 marks.
19. Summarize the importance of enzyme-based processes in improving wine quality and production efficiency.[K] / [CO2]
    Wine production- process- 2 marks, importance of enzyme-based processes- 2 marks.

[4x5 = 20]

Part C-Essay

Answer any 1 questions-Each question carries 10 marks

20. Discuss the role of Golden Rice in addressing vitamin A deficiency and its potential impact on global health. [U] / [CO3]
    GM crops and Golden Rice- introduction- 3 marks, Production of golden rice – 5 marks, impact on global health- 2 marks.
21. How do bioreactors help in scaling up antibiotic production? [U] / [CO4]
22. Bioreactors- introduction- 2 marks, antibiotic production-explain any one example in detail- 8 marks- 8 marks [10x1 = 10]